Cell

The Cancer Genome Atlas: Creating Lasting Value beyond Its Data

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https://doi.org/10.1016/j.cell.2018.03.042

The Cancer Genome Atlas (TCGA) team now presents the Pan-Cancer Atlas, investigating different aspects of cancer biology by analyzing the data generated during the 10+ years of the TCGA project.

The Cancer Genome Atlas (TCGA) is a joint project of the National Cancer Institute (NCI) and the National Human Genome Research Institute (NHGRI). TCGA began in 2006 as a pilot project focused on three cancer types: lung, ovarian, and glioblastoma. Due to the success of the initial efforts, TCGA was reauthorized for a full production phase in 2009. In the following decade, TCGA collected more than 11,000 cases across 33 tumor types and generated a vast, comprehensive dataset describing the molecular changes that occur in cancer (Figure 1). Tissue sample collection and data generation were completed in 2013 and 2016, respectively. Network maker papers, which are integrative cross-platform analyses of TCGA data on individual cancer types have been published for 31 of the tumor types to date (https://cancergenome.nih.gov/publications), with the last two marker papers being prepared for publication in Spring 2018.

It is difficult to overstate the value of the TCGA dataset. Its richness has enabled researchers to catalog specific genomic and molecular changes that occur in cancer, to define a more meaningful taxonomy of cancer types and subtypes, and to even investigate questions that were not imagined at the outset of the project, such as the mining of the data to discover new viruses and other microbial agents. The TCGA marker papers have served as additional resources for understanding the molecular features of these cancers, and a launching pad for individual researchers to deepen the exploration of the data generated.

In the current issues of *Cell*, *Cancer Cell*, *Cell Systems*, *Cell Reports*, and *Immunity*, the TCGA team presents a series of 26 manuscripts investigating different aspects of cancer biology by analyzing the whole bolus of data generated during the 10+ years of the TCGA project. We have dubbed this effort, which was designed as the culmination of the project from a coordinated team perspective, the Pan-Cancer Atlas or PanCan Atlas, for short. The different manuscripts broadly fall into three categories: (1) cell of origin, with tumor aggregation based on a biological system approach (Pan-Gynecological, Pan-Gastrointestinal) or histological subtypes (Pan-Squamous Cell Carcinoma); (2) oncogenic processes across all the tumor types characterized (immunity, mutation, aneuploidy); and (3) surveys of the role different pathways play in the development of different tumor types. As with previous cross-cancer analysis, The PanCan Atlas efforts uncover patterns across related tissue types and provide the scientific community with a wide-ranging overview of different aspects of cancer biology revealed through comprehensive analysis of a large collection of cases. We do not see the PanCan Atlas as the final analysis of the data; but, instead, as another launching pad from which researchers can explore the ensemble of the data generated by TCGA to perform both specialized cancer research, and broader biomedical research.

While the TCGA data and resulting analyses will continue to serve as a valuable resource for scientific exploration by the community, we believe that the greatest impact of TCGA has been its transformation of the way cancer genomics is conducted. Throughout the years, the operation of the TCGA project has produced a number of ancillary results that are impacting biological research beyond the findings derived from the rich TCGA dataset. Examples of these "intangible" results are:

(1) Team Science. TCGA demonstrated that integrative, cross-platform, interdisciplinary investigation made possible by scientific consortia is an extremely valuable and valid model of research. The TCGA project was greatly enhanced by the depth and interconnection of different expertise, as teams of researchers set out to perform analytical tasks that would have otherwise been unattainable by individual efforts. Further, TCGA has demonstrated that consortia participation can be a career-building platform (contrary to the common wisdom that only first authorship can help in attaining senior position in academia). Even with most of the marker papers having only the network banner as a single author, many scientists have built a successful academic career through participation in TCGA. Early career scientists participating in TCGA have received academic credit and recognition, in many cases establishing themselves as the world leader in their field of analysis. Thanks to TCGA (and many other large-scale projects that also follow the model), team science will continue to thrive in biomedical research, benefitting science as well as the scientists.
(2) Genomic Pipelines. Due to the richness of the datasets created,

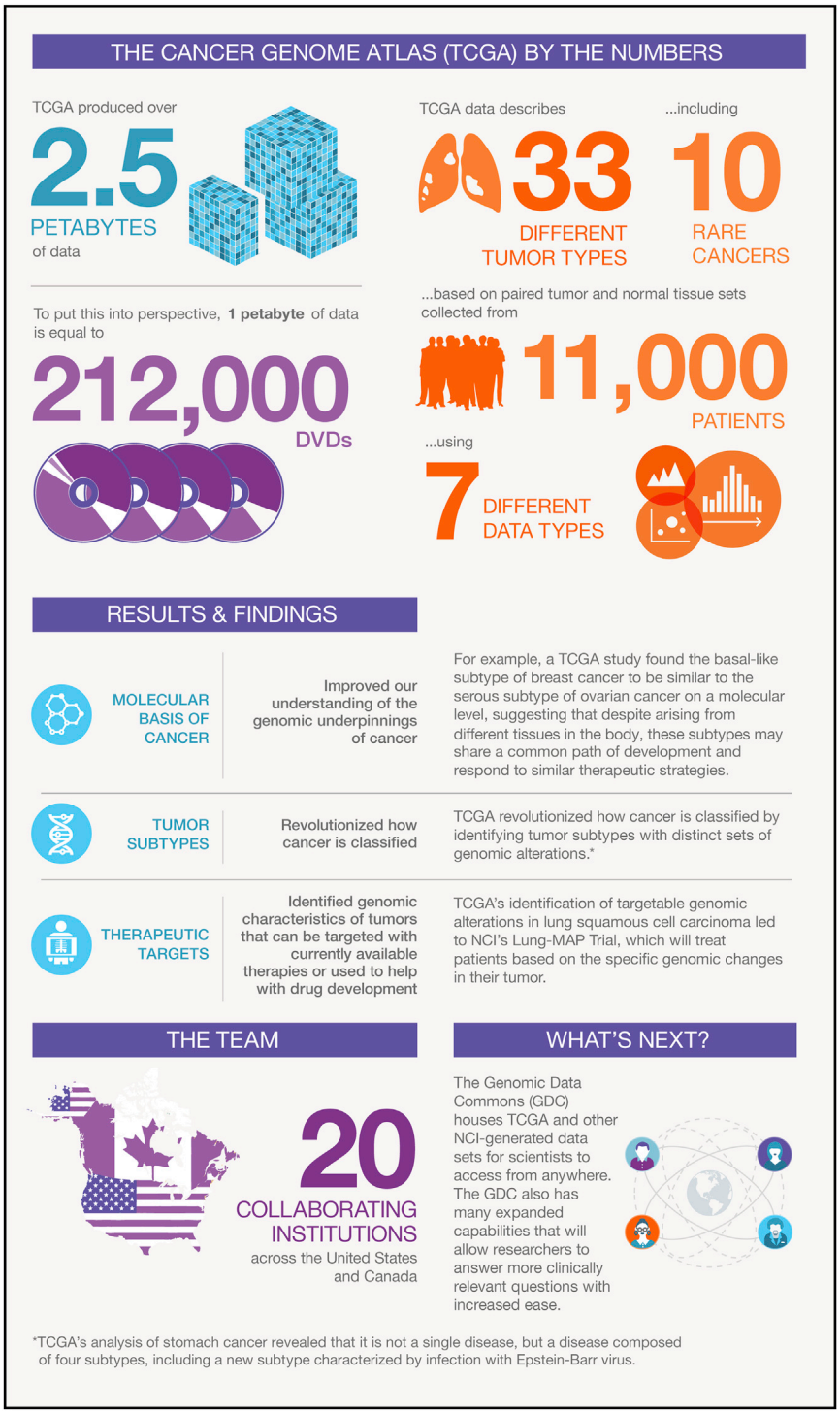


Figure 1. TCGA by the Numbers
An infographic summarizing the scale of the effort and output from the Cancer Genome Atlas (TCGA).

algorithm developers have been attracted to TCGA, creating and honing methods for genomic data analysis, since early days of the project. A number of these developers were not originally familiar with the oncology field. Many of the tools developed in the TCGA setting have become standards in genomic analysis of any kind, helping discovery in all forms of human disease investigation and even biology at large. The necessity of creating data that is consistent and reproducible across thousands of cases, along with the sheer size of the TCGA dataset, forced investigators to innovate in the fields of data science and computational biology. The protocols developed for tissue sample collection and processing, as well as the pipelines for genomic data production and analysis, have become one of the most frequently requested items from the TCGA program. Data are important, but how the data were generated is a vital aspect that many times lurks in the background. With the free access of all TCGA protocols, metadata have come to the main stage, where they truly belongs. Thus, the pipelines, standards, and best practices that TCGA developed have become invaluable resources for the research community, and will continue to serve as the backbone for future genomic research projects.

(3) Sample Acquisition and Study Design. TCGA has been instrumental in highlighting the need for thoroughly annotated, high-quality tumor samples with matched normal controls and informed consent from the patient. When the project began in 2006, the idea of annotated samples with match controls was only truly present in the epidemiology field; thanks in part to the visibility of TCGA, it has come to be at the forefront of multiple areas in biological research. In fact, the NCI now requires that clinical trials follow a set of standard practices to make high-quality tissue samples available for researchers to perform correlative studies. Many of these requirements are taken directly out of the TCGA playbook. These improvements in clinical trial design and organization greatly facilitate the integration of cancer genomics and clinical cancer

research, which can only result in easier translation of molecular discoveries to the bedside.

4) Data Accessibility. One of the main tenets of the TCGA program has been that all the data generated must be accessible as a resource for the scientific community and the general public. TCGA developed and implemented policies where the data were available for public access even before the marker papers describing the characterization of each tumor type was published. Such an openness in sharing was truly innovative at the start of the project. The resulting publications, both within and outside the TCGA network, demonstrated that data sharing is not to be feared, but instead embraced, as it enhances the value of the data by allowing different viewpoints to do the analysis in original and unexpected ways. TCGA was one of the early pioneers of broad data sharing with appropriate privacy protections, leading the way for other models such as the International Cancer Genome Consortium data haring, and the NIH Genomic Data Sharing Policy. Additionally, this desire to allow data use by all interested parties, was the impetus in developing the NCI Genomic Data Commons (GDC). The GDC houses data, not only from TCGA but from a host of genomic projects, uses standardized pipelines to process harmonized data files for cross-project comparisons, and provides access to both unrestricted, open-access, and protected, controlled-access, data. Moreover, the GDC has developed a visualization and analysis interface for the data that obviates the need to download petabytes by allowing users to simply explore the information through a webpage, fully democratizing the data access.

We are proud that TCGA has enabled a significant shift in the way biomedical research is conducted, clearly establishing the model of resource generating projects done in a large team collaborative setting as vital for the elucidation of the complex, multi-dimensional problem of human disease. The PanCan Atlas provides an example of how these efforts can culminate in a set of harmonized data, standardized analysis, and synthesized research findings, that enhance our understanding of cancer biology, and move the field closer toward the goal of improving our ability to diagnose, treat and prevent cancer.